

Short-Horizon Forecasting of the SPY Implied-Volatility Surface

A pre-registered study on Databento OPRA data

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Abstract

Are short-horizon moves of the SPY implied-volatility surface predictable, and does the forecast retain positive vega-attributed edge after observable spread costs? I build one-minute, quality-gated and static-arbitrage-screened volatility-surface states from OPRA consolidated top-of-book quotes (4.4×10^{10} records over the initial 21-session panel, June 2–July 1, 2026, later extended to 64 sessions spanning March 31–July 1), then forecast per-cell implied-volatility changes at 5–60-minute horizons under a purged walk-forward protocol with mechanical leakage probes and shuffled-label negative controls. At the primary 15-minute horizon, ridge regression on a compact set of time- t features improves vega-weighted MSE over the persistence null by 16.7% on the June panel (25.9% unweighted); on the expanded three-month panel its unweighted skill is 26.1%. A GPU sweep of temporal convolutional networks shows a deep sequence model adds essentially nothing: within the tested configurations, receptive fields beyond 15 minutes produced no additional out-of-sample skill. Pre-registered feature ablations reveal that a finding as intuitive as “Greeks help” flips sign between a one-month and a three-month panel. A final symmetric experiment finds that, in the tested one-minute per-cell representation, Lee–Ready-signed prints add no measurable incremental out-of-sample skill beyond quote-based order-flow features at any horizon. Every expectation was registered before the corresponding run; negative results are reported as registered. The economic question is settled by a pre-registered execution-aware replay (Section 6) whose design, endpoints, and verdict taxonomy were locked before any replay computation and which ran exactly once: at the primary 15-minute horizon the forecast does *not* clear the touch (mean entry-cost markout $-\$0.0030$ per share, 95% day-bootstrap CI $[-0.0040, -0.0021]$), even though it genuinely selects favorable repricing; a registered attribution decomposition shows the model prices cell-median dislocations that the tradable contract realizes only fractionally at short horizons. An exploratory horizon sweep, reported with its multiple-comparison caveat, finds the same forecasts clear entry costs decisively at 60 minutes, naming the next pre-registerable hypothesis.

1 Question and setup

The object being forecast is the surface state $IV_t(\text{cell})$: for each cell of a fixed (days-to-expiry $\times$ log-moneyness) grid, the target is the forward change

$$\Delta IV(t, h) = IV(t + h) - IV(t), \quad h \in \{5, 15, 30, 60\} \text{ minutes},$$

*Forecasting results correspond to `qfdev_research` commit `3eba5bf8`; the execution-replay design of Section 6 is locked at commit `613a309f`, pushed before any replay computation; the single replay run used code at `884ff426` and its adjudication is committed at `78c035d4`.

with 15 minutes as the primary horizon. The null model is persistence ($\Delta IV = 0$), the surface staying where it is, which at these horizons is a genuinely hard baseline, not a strawman.

“Good” is defined by what a desk would do with the forecast, not by raw RMSE. The eventual use is relative-value: to first order, the volatility-attributed premium change of an option is approximately $\text{vega}_t \times \Delta IV$ per share (dollar results per standard contract multiply quoted premium changes by the 100-share multiplier). This first-order attribution excludes delta, gamma, theta, and volga effects, hedging, and financing; the execution replay of Section 6 therefore reports a *vega-linearized* edge, never total option P&L. The headline forecasting metric is *vega-weighted* MSE skill versus persistence (defined exactly in Section 3), alongside the per-minute cross-sectional rank information coefficient. The tradeable scope was locked in advance to short-dated wings (expiries within seven days, strikes off the immediate at-the-money point) because that is where both the movement and a realistic edge concentrate.

2 Data and surface construction

The quote source is Databento OPRA consolidated NBBO top-of-book (**cmbp-1**) for all listed SPY options, with Nasdaq **mbp-1** for the underlying. Three nested samples appear in this memo:

Stage	Dates (2026)	Sessions	Purpose
Initial panel	Jun 2 – Jul 1	21	Baselines and first ablations
Expanded panel	Mar 31 – Jul 1	64	Architecture sweep; pre-registered replications
Replay prediction dump	Apr 15 – Jul 1	54 test days	Execution-aware replay (Sec. 6)

Table 1: The three sample stages. The replay dump contains the 7,633,773 out-of-sample prediction rows produced by the walk-forward protocol on the expanded panel; its 54 test days are the expanded panel’s out-of-sample days.

The initial 21-session panel (June 2 to July 1; “the June panel” below) covers roughly 858 GB of raw quotes, 44.05 billion records, verified row-for-row against the vendor’s manifest.

Quotes are compacted into one-minute surface snapshots. Each snapshot fits SVI slices [2] per expiry with quality gating (85.7% of slices trusted on the reference validation day; the rest fall back to direct interpolation), recovers forwards from put–call parity (100% coverage), and writes IV, Black-76 [1] Greeks, and quality flags onto a fixed grid of 9 DTE buckets $\times$ 14 log-moneyness nodes: 126 cells per minute. The June panel used for forecasting is 886,467 cell-minutes.

Every session must pass an automated validation gate before it enters the panel. Among contract-minutes passing the quote-eligibility filters (finite two-sided quotes with $0 \leq \text{bid} \leq \text{ask}$ and a positive mid, the builder’s **ok** flag), the fraction with a solvable IV must be $\geq 90\%$; coverage fractions quoted against *all* raw contract-minutes, including ineligible and malformed quotes, are lower by construction and are not the gated quantity. Further gates: median put–call IV disagreement ≤ 1.5 vol-points, butterfly-negativity (a static-arbitrage smell) below threshold, SVI fit RMSE bounds, and an external sanity check of ATM IV against an independent reference within 3 vol-points. All 21 June sessions passed.

3 Methodology and pre-registration

Models. Five model classes appear throughout: persistence (the $\Delta IV = 0$ null); a constant train-mean baseline (the drift-aware null used by the negative controls); ridge regression ($\alpha = 1$

on standardized features) as the locked reference model; LightGBM as a vega-weighted challenger; and a causal dilated temporal convolutional network (TCN) [4] trained with a vega-weighted Huber loss.

Vega-weighted metric. The headline metric is skill versus persistence under vega-weighted MSE,

$$\text{MSE}_v = \frac{\sum_i w_i (y_i - \hat{y}_i)^2}{\sum_i w_i}, \quad w_i = \text{vega}_i,$$

where vega_i is the raw per-cell Black-76 vega written by the surface builder at decision time t (no cap, floor, or normalization; rows with missing vega are excluded), and skill is $1 - \text{MSE}_{\text{model}} / \text{MSE}_{\text{persistence}}$ computed under the same weights. Weighting always enters *evaluation*; it enters *training* only where stated: LightGBM via per-row sample weights and the TCN via its vega-weighted loss; ridge is trained unweighted.

Cross-sectional rank IC. At each minute with at least 20 in-scope cells, the Spearman rank correlation between predicted and realized ΔIV is computed across cells; the reported figure is the unweighted mean over minutes. Because 15-minute labels are sampled every minute, the raw t -statistics over minutes are overlap-inflated; deflating by $\sqrt{15}$ is quoted alongside where it matters.

Validation protocol. Five-fold expanding-window walk-forward over the minute axis, purged and embargoed by the longest horizon (60 minutes) so overlapping labels cannot leak across the train/test boundary [6]; roughly 600–700k out-of-sample rows per horizon on June alone. Feature standardization is fit on training folds only.

Leakage controls. Two mechanical probes run against the feature builder itself: truncation invariance (features computed on a panel cut at time t must be bitwise identical to the full-panel build) and future-input perturbation (corrupting all raw inputs after t must leave features at or before t unchanged). The probes were validated by injecting a deliberate future-IV feature, which they caught. In addition, shuffled-label negative controls were run for every model class: within each walk-forward fold, *training* labels are permuted after the purge/embargo (test labels intact) and the models retrained. Because the target has non-zero mean over long windows, shuffled models are judged against the constant train-mean baseline rather than the zero null; otherwise target drift masquerades as leakage. Measured feature-driven excess collapses to $\leq +0.004$ against a 0.02 tolerance (versus +0.17 to +0.31 on real labels), and feature importances go flat.

Significance tests. Reported Diebold–Mariano [5] p -values test the per-minute loss-differential series between two arms scored on identical row sets (cross-sectional losses aggregated within the metric’s weighting), with a Newey–West long-run variance at lag equal to the horizon length in minutes to absorb label overlap. They are computed per (model, horizon, metric) cell and are *not* adjusted for multiplicity across the grid; they serve as supporting diagnostics. The primary evidence standard everywhere is the registered effect threshold with the sign holding in every walk-forward fold and every calendar month.

Pre-registration. Before each experiment round, the expected outcome, the primary endpoints, and the decision rules were written down and committed. Example: Greeks become default-on for a model class only if they add at least +1.0 percentage point of vega-weighted skill on at least two

horizons *with the sign holding in every walk-forward fold*. The point is to make the negative results as legible as the positive ones.

4 Results

4.1 Baselines beat persistence; the signal is mostly linear

Walk-forward skill versus persistence on the June panel:

Skill vs. persistence	5m	15m	30m	60m
Ridge, plain MSE	+25.2%	+25.9%	+24.6%	+30.9%
LightGBM, plain MSE	+21.5%	+16.7%	+18.3%	+24.9%
Ridge, vega-weighted	+19.6%	+16.7%	+8.3%	+8.0%
LightGBM, vega-weighted	+17.8%	+11.9%	+7.0%	+1.3%

Table 2: Out-of-sample skill, 21-session June panel, 5-fold purged walk-forward. LightGBM rows use vega sample-weighting in training. Values are point estimates; day-clustered uncertainty is currently reported only for the registered adjudications (fold ranges, DM diagnostics) and the replay’s day bootstrap; extending day-bootstrap intervals to this table is queued work (Section 7).

Ridge matches or beats LightGBM on plain MSE at every horizon: the signal is largely linear in this feature set. The weighting matters more than the model: an unweighted LightGBM is *worse than persistence* at the 15-minute horizon once evaluated vega-weighted (-1.3%), and simply training with vega sample-weights swings it to $+11.9\%$, a 13-point difference from an evaluation choice, before any modeling work. Directional accuracy decays from 58% at 5 minutes to 53% at 60 minutes. On short-DTE wings, the locked reference model achieves a mean per-minute cross-sectional rank IC of $+0.40$ (raw $t \approx 116$ over minutes, ≈ 30 after $\sqrt{15}$ overlap deflation; 91% of minutes positive) with a vega-weighted calibration slope of 0.992 (vega-WLS of realized on predicted through the origin), so predicted magnitudes can be taken at face value in the tradeable scope. The adjacency of surface cells makes the per-minute cross-section dependent, so the IC’s minute count overstates its effective sample size; the fold- and month-consistency standards of Section 3 are the load-bearing evidence.

4.2 A negative result on architecture

An 11-run TCN sweep on the full 64-session span (single RTX 4060 Ti, ~ 15 h wall) tested whether a deep sequence model finds anything the cheap models miss. It does not: TCN skill ties ridge and LightGBM at every horizon: at 15 minutes the *plain-MSE* skill fractions are ridge 0.261, LightGBM 0.251, TCN 0.248 (the sweep’s summary metric is unweighted skill versus persistence). A receptive-field ablation is the sharpest summary: growing the TCN’s context from 3 to 7 minutes helps, and within the tested configurations, increasing the receptive field beyond 15 minutes produced no additional out-of-sample skill. That bounds the incremental value of longer memory for this model class and representation; it does not establish that no longer-memory information exists. In a capacity sweep, additional dropout did not improve out-of-sample performance (dropout 0 was optimal), suggesting insufficient regularization was not the limiting factor in the tested capacity range. The feature set does the work, not the architecture; the next lever is cross-sectional structure, not a bigger network.

4.3 Feature ablations: importance is not generalization

On June, adding Black-76 Greeks as features *hurt* LightGBM by 4 to 11 skill points depending on horizon (worst at 30m: -10.6pp), while leaving ridge indifferent, even though Greeks captured 8.6% of the model’s split gain at that same horizon. A plausible interpretation was that gamma, theta and vega are near-deterministic transforms of features already present, and that the redundancy encouraged unstable tree splits that spend themselves on out-of-sample variance. The expanded-panel replication in Section 5 was designed to test exactly that interpretation. Quote-based microstructure features (order-flow imbalance, quote imbalance) added $+1.5\text{pp}$ for LightGBM at 30/60m and nothing below that. Both findings were treated as hypotheses to re-test, not conclusions.

5 What changed between rounds, and why

The June ablations were re-run as a pre-registered 2×2 (Greeks $\times$ microstructure) on the three-month panel (the five model classes of Section 3 $\times$ 4 horizons $\times$ 4 arms), with predictions and decision rules committed before launch. The registered expectations were: ridge flat under Greeks ($|\Delta| < 0.5\text{pp}$), LightGBM hurt by $\geq 1\text{pp}$ on ≥ 2 horizons (replicating June), TCN approximately flat, and no Greeks $\times$ micro interaction.

The replication broke the June story in an instructive way. All contrasts below are computed on identical row sets, with per-fold and per-calendar-month deltas and Diebold–Mariano tests as described in Section 3. The LightGBM Greeks harm, the largest and most intuitive June effect, did *not* replicate: at the 30-minute replication target the sign flipped from -10.6pp to $+0.7\text{pp}$ (all five folds positive, DM $p < 10^{-4}$), with $+0.3$ to $+0.7\text{pp}$ across the other horizons. Ridge, registered as flat, instead gained $+1.9\text{pp}$ of vega-weighted skill at 30 minutes and $+4.8\text{pp}$ at 60 from Greeks, with the sign holding in every walk-forward fold and every calendar month, so under the rule fixed in advance, Greeks flipped default-on for the ridge class (and only there: no other model class met the threshold; a fold-consistent TCN cell at 30 minutes had no vega-weighted counterpart and was discarded under the registered stability standard). The registered mechanism story, that the June harm was specific to greedy split selection, failed its own preconditions and is reported as wrong. The microstructure edge shrank from $+1.5\text{pp}$ to $\sim +0.3\text{pp}$ at 30/60 minutes; its registered replication target (LightGBM 60m $\Delta > 0$) formally held ($+0.31\text{pp}$, all folds positive, DM $p = 0.026$) but attenuated fivefold. At the 15-minute story horizon every effect was null. The plain reading: one month is one regime, and feature-ablation signs measured on it do not transfer. The pre-registration is what makes this a finding rather than an embarrassment: the June conclusions were recorded, the re-test criteria were fixed in advance, and the reversal is reportable exactly because nothing was re-fit after seeing the answer.

A symmetric test of trade flow

The quote-based order-flow features raise an obvious question: do actual *prints* carry anything the book does not? A second pre-registered 2×2 (microstructure $\times$ signed trade flow) added Lee–Ready [3] signed volume per cell from OPRA trades, engineered with exactly the same 30-minute temporal treatment as the quote-pressure features so the comparison could not be confounded by representation, and framed two competing hypotheses in advance: prints add information beyond posted pressure (H-new) versus quotes impound trades within the minute (H-span). Results consistently favored H-span over H-new within the registered representation. The primary endpoint (LightGBM skill change from flow given the quote features, 60-minute horizon) was -0.07pp (DM $p = 0.46$); the largest ridge cell was $+0.07\text{pp}$; and the substitution contrast (flow alone versus

neither) was equally null, so there was no flow credit for the quote features to steal. Two TCN cells produced large point estimates (+3.0pp at 30m vega-weighted, DM $p = 0.004$) with per-fold deltas spanning -1.1 to $+11.5$ pp, exactly the multiple-comparisons bait the registered fold-consistency standard exists to discard. Two caveats are owned rather than hidden: Lee–Ready misclassifies option trades (multi-leg packages especially), and per-cell signed volume conflates directional with volatility demand; the strongest untested variant, chain-level vega-weighted net flow, is named in Section 7 rather than quietly run.

The replication round also doubled as an engineering exhibit: the two overlapping arms were recomputed on a second machine (Windows, pinned to the same library versions) and all 64 recomputed ridge and LightGBM summary values matched the Linux results bit-for-bit. (The claim is scoped to those table values; row-level prediction files were not hashed.)

6 Execution-aware replay: does the edge survive the spread?

Vega-weighted forecast skill is not a trading result. This section closes that gap with a pre-registered execution replay that maps each in-scope cell forecast to an actual quoted contract and charges observable spread costs. The full design, endpoints, falsifiable expectation bands, and verdict taxonomy below were committed (613a309f) *before any replay computation existed*; the replay runs exactly once against them.

6.1 Registered design (summary)

For each (minute, cell) a single representative OTM-side contract is chosen on time- t information only (nearest log-moneyness to the node among quote-valid contracts, tie-broken by relative spread); the exit is the *same* contract at the exact quote h minutes later, with no forward fill; a missing exit is counted as unfillable and excluded from markout, never silently filled. Vega is the per-contract Black-76 vega at t . A trade fires iff $|\text{vega}_t \cdot \lambda \cdot \widehat{\Delta IV}| > \theta \cdot \text{half-spread}_t + s$, with $\theta = 1$, $\lambda = 1$, $s = 0$ primary and $\theta \in \{1, 1.5, 2\}$, slippage $s \in \{0, 0.005, 0.01\}$ \$/share, and a walk-forward-calibrated λ arm as sensitivities. Three outcome quantities are kept distinct: the gross vega-linearized edge $\text{sign}(\widehat{\Delta IV}) \cdot \text{vega}_t \cdot (IV_{t+h} - IV_t)$; the *entry-cost-adjusted markout* (gross minus half-spread at entry, marked to $t+h$ mid), the statistical primary; and the *round-trip-touch net* (additionally minus the exit half-spread), which drives the economic verdict. Scope is the registered tradeable region ($\text{DTE} \leq 7$, off-ATM); out-of-scope cells are evaluated hypothetically and reported separately. Inference is a day-level bootstrap ($N = 10,000$) over the 54 test days. A greedy per-cell cooldown variant provides a non-overlapping replay; its cumulative output is a cumulative one-contract non-overlapping edge, *not* a portfolio equity curve, and is reported with concurrency diagnostics. A delayed-entry arm (decide at t , enter at the $t+1$ -minute touch) bounds the latency objection.

6.2 Primary endpoint and decision rule

The registered primary endpoint is the pooled mean entry-cost-adjusted vega-linearized markout per fillable gated trade (ridge, 15-minute horizon, in-scope, $\theta = 1$, $\lambda = 1$, $s = 0$) with a 95% day-bootstrap confidence interval.

The endpoint is invalidated (reported as “diagnostics only”, no headline claim) if gated exit unfillability exceeds 5% overall or is more than twice the overall rate in the top predicted-edge decile; selective unfillability where the model claims the most edge would make the markout uninterpretable.

6.3 Registered verdict taxonomy

The prose conclusion is constrained to one of three registered verdicts:

- **Survives observable entry costs.** Primary CI entirely above zero, round-trip-touch point estimate ≥ 0 with its sign preserved in the non-overlapping variant, validity condition intact.
- **Markout survives entry costs; round-trip execution not demonstrated.** Primary CI above zero, but the round-trip criterion fails.
- **Does not survive costs.** Primary CI straddling or below zero, or the endpoint invalidated by selective missing exits, reported as such: 15-minute surface forecasts of this quality do not clear the touch.

Six falsifiable expectation bands (gate pass rate, gross selection monotonicity in θ , the honest spread-survival band, LightGBM overconfidence as a trading statement, out-of-scope deterioration, and non-overlap sign preservation) were registered alongside; the strong result (round-trip positive with CI above zero) is registered as a *surprise*, not the expectation.

6.4 Results

One amendment was registered between the design commit and the run: the registered one-day smoke test fired its attribution gate (on the smoke day the tradable contract’s ΔIV realized only $\sim 0.2\times$ the cell-median target the model predicts), so a pre-specified attribution decomposition (floating cell median vs. fixed-membership median vs. single contract) was added to the endpoint set (884ff426) *before the full run*, with the primary endpoint and every expectation band unchanged. The replay then ran exactly once: 54 test days, three models, all registered arms.

Gates and validity. The mechanical self-test and both lookahead probes pass. One registered bound breached marginally and is disclosed rather than amended: 2.4–2.6% of predictions per horizon matched no tradable contract, against a registered 2% bound. Every such case is a far-wing cell whose node contained no quote-valid OTM-side contract, a cell on which the replay’s instrument is not defined; there were zero join failures. The validity condition holds on both clauses: gated exit unfillability is 1.53% ($< 5\%$) and *falls* with predicted edge (0.37% in the top decile), so missing exits do not concentrate where the model claims the most edge.

Primary endpoint. Ridge, 15-minute horizon, in scope, $\theta = 1$, $\lambda = 1$, $s = 0$: mean entry-cost-adjusted markout per fillable trade **−\$0.0030** per share, 95% day-bootstrap CI $[-0.0040, -0.0021]$, over 509,946 trades on 54 of 54 days (the equal-day-weighted mean, $-\$0.0029$, agrees); round-trip-touch net $-\$0.0098$ $[-0.0108, -0.0088]$. The verdict is the registered third: **15-minute surface forecasts of this quality do not survive costs: they do not clear the touch.**

Registered expectations. Of the six bands, three held and three were rejected, and the rejections are the informative part. *Supported:* gross selection works (conditional mean gross vega-linearized edge $+0.0033$ at $\theta = 1$, rising monotonically to $+0.0040$ at $\theta = 2$, same pattern for all three models; the failure is cost-sized, not signal-sized); out-of-scope cells are worse for the registered model (conditional gross $+0.0023$ vs. $+0.0033$ in scope); the non-overlapping replay preserves every sign (trade-count ratio 0.086, primary $-0.0030 \rightarrow -0.0038$). *Rejected:* the gate pass rate came out at 71.8% against a registered $[2, 30]\%$ band: the gate believes the cell-median move, while

the fill gets the contract (the mechanism the decomposition below quantifies); LightGBM did *not* over-trade relative to ridge (pass rate 55.9%, per-trade markout -0.0028); and the markout did not survive entry costs (the primary above).

Attribution: what the model predicts vs. what a contract realizes. The amended decomposition separates membership churn from single-contract attenuation (Table 3). Membership churn is a roughly constant haircut: the fixed-membership cell median tracks the target at $\text{corr} \approx 0.7$ at every horizon. The horizon-dependent loss is per-contract: at 5–15 minutes the cell-median ΔIV the model forecasts is dominated by quote-bounce of the cell’s wide members; that bounce is mean-reverting and genuinely *predictable*, but not realizable on the tight tradable contract (regression slope of contract on fixed-median ΔIV : 0.09 at 5m, 0.25 at 15m). By 60 minutes the contract realizes roughly $0.7\times$ the cell move.

h	$\text{corr}(y, \text{fixed})$	$\text{corr}(\text{fixed}, \text{contract})$	slope (contract on fixed)	$\text{corr}(y, \text{contract})$
5m	0.74	0.28	0.09	0.13
15m	0.70	0.47	0.25	0.26
30m	0.70	0.64	0.44	0.41
60m	0.73	0.81	0.68	0.60

Table 3: Registered attribution decomposition, in scope. y is the floating cell-median target the model is trained on; “fixed” freezes the cell’s membership at t ; “contract” is the single tradable contract the replay fills. Cell-level predictability survives at all horizons; its tradable share grows with horizon.

Exploratory: the economics are monotone in horizon. With the explicit multiple-comparison caveat that the registered primary was 15 minutes, the per-trade economics improve with horizon exactly as the decomposition predicts: entry-cost markout -0.0058 at 5m, -0.0030 at 15m, $+0.0009$ at 30m (CI straddling zero), and **$+0.0083$** at 60m with CI $[+0.0043, +0.0117]$ entirely positive; the 60m round-trip-touch net is $+0.0015$ $[-0.0026, +0.0049]$: breakeven, not a win. The 60m entry-cost result survives the registered honesty arms: $+0.0078$ under one-minute delayed entry and $+0.0077$ under non-overlapping sampling; on the slippage grid it remains positive through $s = \$0.005$, whereas at 15m the same s is immediately fatal (-0.0076). This is reported as what it is: not a claimed result, but the precise, pre-registerable hypothesis the registered run points to, namely to confirm the 60-minute horizon as primary on new data.

7 Limitations and next steps

- **One underlier, one quarter.** Everything here is SPY, March 31 – July 1, 2026 (approximately Q2). The regime-sensitivity result in Section 5 is itself the argument for not extrapolating further than the data supports.
- **Execution-aware markout, not a portfolio backtest.** The replay of Section 6 maps cell forecasts to actual option quotes and charges observable spread costs, but its primary outcome is a vega-linearized markout. It does not model delta hedging, nonlinear Greeks, queue position, displayed-size constraints, market impact, portfolio capital, or asynchronous latency (the delayed-entry arm bounds, but does not remove, the latency objection). The same-bar touch assumption is an execution benchmark, not a live-fill claim. Source quotes

may have been forward-filled by up to five minutes in the surface builder, and the slim contract extract does not carry a per-row staleness flag.

- **Point estimates in the forecasting tables.** The headline skill tables report point estimates; day-clustered uncertainty currently exists only where the registered adjudications required it (fold ranges, per-month splits, DM diagnostics) and in the replay’s day bootstrap. Extending day-bootstrap intervals to Table 2 is queued.
- **Surface gates are static-arb smells, not proofs.** The validator checks butterfly negativity and put–call consistency; a full calendar/butterfly no-arbitrage repair across the grid is open work.
- **The flow null is a null for one representation.** Per-cell, one-minute, Lee–Ready-signed volume with window aggregates adds nothing; that bounds the *average* contribution, not a localized one. The named untested variants: chain-level vega-weighted net flow (attacks the dealer-inventory channel directly, endpoint: LightGBM vega-weighted skill at 30/60m), skill conditioned on cell-minutes with nonzero window flow (attacks sparsity dilution), and multi-leg print exclusion (attacks sign misclassification).
- **Next:** pre-register and confirm the 60-minute horizon as primary on new data (the one hypothesis the replay’s exploratory sweep names; Section 6.4); cross-sectional feature structure (the direction the architecture sweep points to); per-model feature defaults in code (the ridge/Greeks flip is adjudicated but ridge and LightGBM currently share one feature matrix).

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